

Artificial Intelligence (AI2002)

Sessional-II Exam

Date: April 9th 2025

Course Instructor(s)

Dr. Hashim Yasin, Dr. Rabia Maqsood, Ms. Aqsa
Younas, Mr. Ali Raza, Ms. Mahzaib Younas

Total Time (Hrs.): **1**
Total Marks: **55**
Total Questions: **5**

Roll no

Section

Student Signature

Vetted by: _____ Signature: _____

Attempt all the questions.

CLO#	1	2	1	1	3
Q#	1	2	3	4	5
Total Marks	08	16	08	15	08
Obtained Marks					

[C1]: To introduce the notion of intelligence and the so called artificiality associated with it, and how these can be modelled in computational system

Question No 1: Fill in the blanks with the correct terms to complete the statement. Cutting and Overwriting will lead to ZERO marks. [8 Marks]

- Pruning allows us to ignore portions of the search tree that make no difference to the final choice, and _____ functions allow us to approximate the true utility of a state without doing a complete search.
- In game parlance, we say that this tree is one move deep, consisting of two half-moves, each of which is called _____.
- _____ move is allowed to escape plateaus in hill climbing?
- The formal definition of entailment is _____

National University of Computer and Emerging Sciences

Chiniot-Faisalabad Campus

5. If variable is left with no more values, then AC-3 (constraint propagation) can immediately return _____.
6. The term backtracking search is used for a depth-first search that chooses values for BACKTRACKING SEARCH one variable at a time and backtracks when a variable has _____ to assign.
7. If the heuristic function is the number of attacking pairs in the 4-queens problem, then write the global minimum for the hill-climbing search. _____
8. Path consistency tightens the binary constraints by using implicit constraints that are inferred by looking at _____ of variables.

[C2]: Knowledge of CSP and ability to apply its techniques practically.

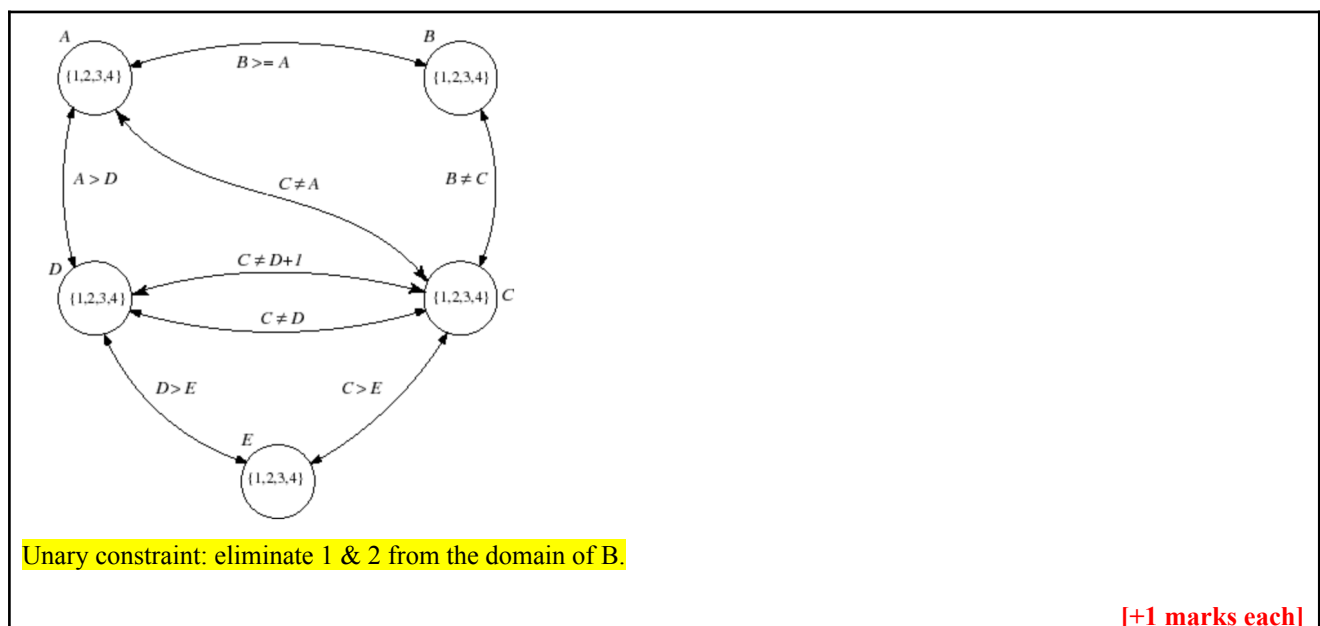
Constraint Specification Problem

[2+8+2+2+2 = 16 Marks]

Question No 02:

Consider a CSP with variables: A, B, C, D, and E, where the domain of each variable is {1, 2, 3, 4} and the constraints are: $A > D$, $D > E$, $C \neq A$, $C > E$, $C \neq D$, $B \geq A$, $B \neq C$, $B > 2$, and $C \neq D + 1$.

- a. Construct the constraint graph that represents the above problem and show all the constraints. Make the graph node consistent if required.



- b. Before doing any value assignment, enforce arc consistency using the AC-3 algorithm. Fill the table below to show all the inconsistent arcs you found, and the value(s) removed to make it consistent.

Arc (var1, var2)	Constraint (var1 * var2) [* is the relation, e.g., var1 > var2]	Value(s) removed
D, E	D > E	D=1 [+1]
E, D	D > E	E=4 [+1]

National University of Computer and Emerging Sciences
Chiniot-Faisalabad Campus

C, E	C > E	C=1 [+1]
D, A	A > D	D=4 [+1]
A, D	A > D	A=1, A=2 [+1]
B, A	B >= A	B=1, B=2 [+1]
E, D	D > E	E=3 [+1]
		[+1] on all correct

c. Write the remaining values for each variable. [+0.5x4 = 2]

A	B	C	D	E
3, 4	3, 4	2, 3, 4	2, 3	1, 2

d. Now encircle the variable that will be picked next by the Minimum Remaining Value (MRV) heuristic?
Pick the right-most variable in case of a tie.

[+2 marks]

A	B	C	D	E
---	---	---	---	---

e. For the above variable, what value will be assigned using the Least Constraining Value (LCV) heuristic?
Pick the right-most value in case of a tie.

E = 1
[+2 marks]

C1: To introduce the notion of intelligence and the so called artificiality associated with it, and how these can be modelled in computational systems.

Beyond Classical Searches

[6 + 2

= 8 Marks]

Question No 03:

National University of Computer and Emerging Sciences

Chiniot-Faisalabad Campus

Part A: In **Simulated Annealing**, assume that the current state's evaluation function value is 15 and the current temperature is $T = 10$. Consider the following randomly selected successor states and decides the move or compute the probability of moving to the successor state,

- i. The selected successor state's value is 6?

- ii. The selected successor state's value is 10?

- iii. The selected successor state's value is 15?

- iv. The selected successor state's value is 25?

Part b: Prove/Justify each of the following statements with suitable assumptions, or give a counterexample: (2)

Local beam search

- i. is the Breadth-first search.

- ii. is the Hill-Climbing search.

Simulated Annealing:

- iii. perform like Hill-Climbing search if the temperature, T , becomes very high.

[C1]: To introduce the notion of intelligence and the so called artificiality associated with it, and how these can be modelled in computational systems.

Adversarial Searches

[4+4+4+3 = 15

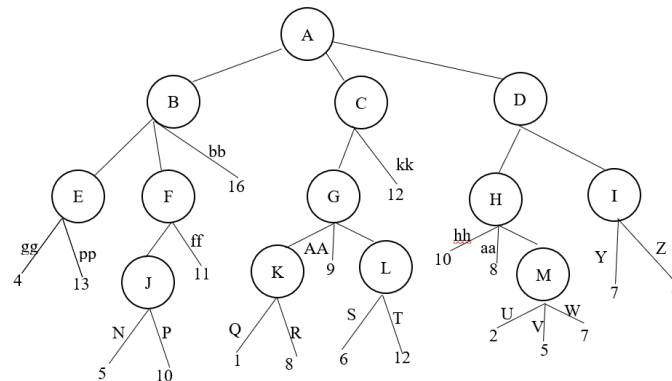
Marks]

Question No 04: Consider a Game tree below and assuming both rational agents are playing the game with **MAX agent** is having a first turn. Use Alpha beta pruning to find optimal path for MAX agent.

National University of Computer and Emerging Sciences

Chiniot-Faisalabad Campus

As we know ordering affects the efficiency of Pruning, you are required to prune this tree in both left to right and right to left order.



a) Write names of all nodes in left to right order which will get pruned

b) Write names of all nodes in right to left order which will get pruned

a) Write complete optimal paths for MAX Agent in both L->R and R->L Order, starting from node A

b) How Ordering affects the efficiency of pruning? Which order is more efficient?

[C3]: To introduce students with computational intelligence theory and techniques used to design and develop intelligent systems.

Knowledge Based Reasoning
Marks]

[6 + 2 = 8

Question No 05:

Part A: Decide whether each of the following sentences is valid, unsatisfiable, or neither. Verify your answers using the equivalence rules. Justify your answer.

[1 Mark of statement and 1 for justification]

i. $(A \leftrightarrow B) \wedge (\sim A \vee B)$

National University of Computer and Emerging Sciences

Chiniot-Faisalabad Campus

A	B	$\sim A$	$(A \leftrightarrow B)$	$(\sim A \vee B)$	$(A \leftrightarrow B) \wedge (\sim A \vee B)$
T	T	F	T	T	T
T	F	F	F	F	F
F	T	T	F	T	F
F	F	T	T	T	T

Not valid, or not unsatisfiable, it is neither or satisfiable.

i. $((y \wedge \Rightarrow P) \wedge (y \Rightarrow \neg) \wedge (\Rightarrow \neg y) \wedge y$

since there are no values of y, P which makes the sentence false. **Unsatisfiability**, since there are no values of y, P which makes the sentence true.

ii. $((\text{Smoke} \wedge \text{Heat}) \Rightarrow \text{Fire}) \Leftrightarrow ((\text{Smoke} \Rightarrow \text{Fire}) \vee (\text{Heat} \Rightarrow \text{Fire}))$

This is valid. You can show this validity by constructing truth tables for the two sides of logical equivalence $\Leftrightarrow$ and verifying this equivalence (by showing that for any assignment of truth values to the variables Smoke, Heat, and Fire, the two formulas are either both true or both false).

Part b: Which of the following are correct. (Give Logical Reason)

[0.5 Mark of statement and 0.5 for justification]

i. $(A \vee B) \wedge (\neg C \vee \neg D \vee E) \models (A \vee B)$

is true; removing a conjunct only allows more models.

he conjunction rule states that if $(P \wedge Q)$ is true, then (P) must also be true. Since $(A \vee B)$ is part of the conjunction, it must always hold.

ii. $(A \vee B) \wedge (\neg C \vee \neg D \vee E) \models (A \vee B) \wedge (\neg D \vee E)$

is false; removing a disjunct allows fewer models.

No. **Logical Reason:** The left-hand side contains $(\neg C \vee \neg D \vee E)$, but the right-hand side requires $(\neg D \vee E)$ specifically. Since $(\sim C)$ is present in the left-hand side but not necessarily in the right-hand side, the entailment does not always hold.